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Mini Project Report on

"IOT - MILITARY SURVEILLANCE SYSTEM"

Submitted in partial fulfillment of the requirements for the
First Semester of the Bachelor of Engineering Degree, towards the completion of the Mini
Project under the Innovation & Design Thinking Laboratory, Department of Basic
Sciences.

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CERTIFICATE

This is to certify that the mini project entitled “Military Surveillance System” has been successfully carried out by Aashish Awasiti (1CR25CS001), Abdul Malik Khan (1CR25CS002), A M Suma (1CR25EC001), and Aakriti Kumari (1CR25ES002), bonafide students of CMR Institute of Technology.

The project is submitted in partial fulfillment of the requirements for the First Semester of the Bachelor of Engineering Degree under the Innovation & Design Thinking Laboratory, Department of Basic Sciences.

It is further certified that all corrections and suggestions made during the Internal Assessment have been duly incorporated. The project report has been reviewed and approved as per the academic requirements prescribed for the said degree.

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Abstract

This project presents the design and development of a military surveillance system integrating hardware and software for perimeter security. The system uses an Arduino microcontroller connected to a thermal sensor for human detection and an ultrasonic sensor for distance measurement, both mounted on a rotating servo motor for wider coverage. Upon detecting a human intruder within the defined perimeter, the system triggers an LED alert, displays a warning message on an OLED screen, and sends real-time data to a laptop dashboard. The software interface provides role-based access for civilians, army authorities, and rescue teams, displaying shelter locations, safe routes, emergency services, army camp coordinates, and missing person reports. The system operates effectively in total darkness and reduces false alarms through passive thermal sensing. The prototype successfully demonstrates real-time intruder detection, alert generation, and multi-role information dissemination, offering a low-cost solution for border surveillance and civilian disaster management.

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Chapter 1

1.1 Introduction

In modern warfare, surveillance and early threat detection play a critical role in ensuring the safety of soldiers and protecting strategic locations. Traditional monitoring methods such as manual patrolling, searchlights, and night-vision observation often expose soldiers to danger and rely heavily on human alertness. Environmental conditions like darkness, fog, rain, and rough terrain further reduce visibility and increase the chances of enemy infiltration. To overcome these limitations, advanced military surveillance systems are being developed using thermal imaging, motion sensing, and Internet of Things (IoT) technologies. These systems provide continuous monitoring, automatic threat detection, and real-time alerts without requiring soldiers to expose themselves to enemy observation. The proposed project, IoT-Based Military Surveillance System, is a low-cost, autonomous early-warning system designed to detect human movement using heat and motion sensors. Once suspicious activity is detected, the system immediately alerts soldiers through visual, audio, and display signals while keeping their position concealed. The system improves battlefield awareness, reduces human fatigue, and enhances border security operations during both day and night conditions.

1.2 Brief History of Military Surveillance Systems

Military surveillance has evolved significantly over the years with advancements in science and technology. In ancient times, surveillance mainly relied on human observation through watchtowers, guards, and signal systems such as smoke or fire alerts. During the World Wars, countries began using binoculars, radar systems, and reconnaissance aircraft to monitor enemy movements over long distances. These technologies improved battlefield intelligence but still required significant human involvement. With the development of electronics and communication systems in the late twentieth century, surveillance methods became more advanced. Infrared sensors, thermal imaging cameras, and motion detectors were introduced to detect enemy presence even in darkness or harsh weather conditions. Later, the rise of digital communication and embedded systems enabled remote monitoring and automated security systems. In recent years, IoT technology, wireless sensor networks, drones, and artificial intelligence have transformed military

surveillance into a highly automated and intelligent process. Modern systems are now capable of real-time monitoring, data transmission, threat analysis, and autonomous alert generation, greatly improving military preparedness and soldier safety.

1.3 Modern Military Surveillance Systems

Modern military surveillance systems are designed to provide continuous monitoring, rapid threat detection, and secure communication without requiring constant human supervision. These systems use advanced technologies such as thermal cameras, infrared sensors, motion detectors, drones, GPS modules, and wireless communication networks to observe sensitive areas effectively. Thermal imaging technology is particularly important because it detects heat signatures emitted by humans, animals, or vehicles, allowing surveillance even in complete darkness, fog, or smoke.

Today's surveillance systems are often integrated with IoT platforms that enable sensors and devices to communicate with each other in real time. Data collected from sensors can be processed automatically and transmitted instantly to soldiers or command centers. Some advanced systems also use artificial intelligence and machine learning algorithms to identify threats, distinguish human movement from environmental disturbances, and reduce false alarms. These modern technologies enhance battlefield awareness, minimize response time, and reduce the dependence on manual monitoring. However, many existing systems are expensive, power-consuming, and difficult to deploy in remote or harsh terrains, creating a need for simpler and more affordable alternatives.

Chapter 2

2.1 About the Project

Military personnel stationed at borders and remote outposts face serious challenges in detecting enemy infiltration at night and in adverse environmental conditions. Traditional surveillance methods, such as searchlights and manual observation, expose soldiers to enemy attacks because they reveal their location and require continuous physical monitoring. Even night-vision devices have limitations, as they may not function effectively in fog, low visibility, or rugged terrain, and they contribute to operator fatigue over long periods of surveillance duty.

There is therefore a need for a low-cost, autonomous, and reliable surveillance system capable of detecting human presence without exposing soldiers to danger. The system should continuously monitor the surrounding area using thermal and motion detection technologies and provide instant alerts through audio, visual, and display signals. Such a solution would improve battlefield awareness, reduce human error, enhance soldier safety, and provide efficient surveillance in a challenging environment.

2.2. Problem Statement

Designing an effective IoT-based military surveillance system involves several technical and operational challenges. The system must accurately detect human presence while minimizing false alarms caused by animals, environmental noise, or changing weather conditions. Since the surveillance unit is intended for remote battlefield deployment, it should operate with low power consumption and maintain reliable performance in harsh environments such as darkness, fog, rain, and mountainous terrain. Another major challenge is ensuring that the system alerts soldiers without revealing their position to hostile forces. The communication mechanism must therefore be secure, silent when necessary, and capable of delivering clear warnings in real time. Additionally, the system should be cost-effective, compact, portable, and easy to deploy so that it can be used efficiently in remote border areas. Achieving high detection accuracy, fast response time, durability, and autonomous operation within a limited budget forms the core challenge of this project.

Chapter 3

3.1 Design

The system consists of a rotating sensor platform mounted on a servo motor. The ultrasonic sensor measures the distance from. The Arduino Uno processes all sensor data. Upon human detection, the OLED displays an alert message, the LEDs turn on, and data is sent via serial to the laptop dashboard. The dashboard provides three role-based interfaces: Civilian (shelters, safe routes, emergency services), Authority (army camp locations, supplies, alerts), and Rescue (missing persons, army contacts).

3.2 Description of the approach to solve the problem

The problem of undetected intrusions at night is solved using passive thermal sensing, which works in total darkness and ignores non-human objects. The rotating motor provides 180° coverage without requiring multiple sensors. The dual-sensor approach (thermal + sonic) confirms human presence before triggering an alarm, reducing false positives. The role-based dashboard solves the problem of information overload by showing only relevant data to each user type. Civilians get evacuation guidance, authorities get tactical data, and rescue teams get coordination tools.

3.3. Steps to implement the surveillance system

1. Assemble the circuit: Connect the sonic sensor to digital pins (Trig pin 9, Echo pin 10), servo motor to pin 6, OLED to I2C, and LED to pin 13.
2. Upload Arduino code to read temperature and distance continuously.
3. Set detection logic
5. On alert: Turn on the LED, display the message on the OLED

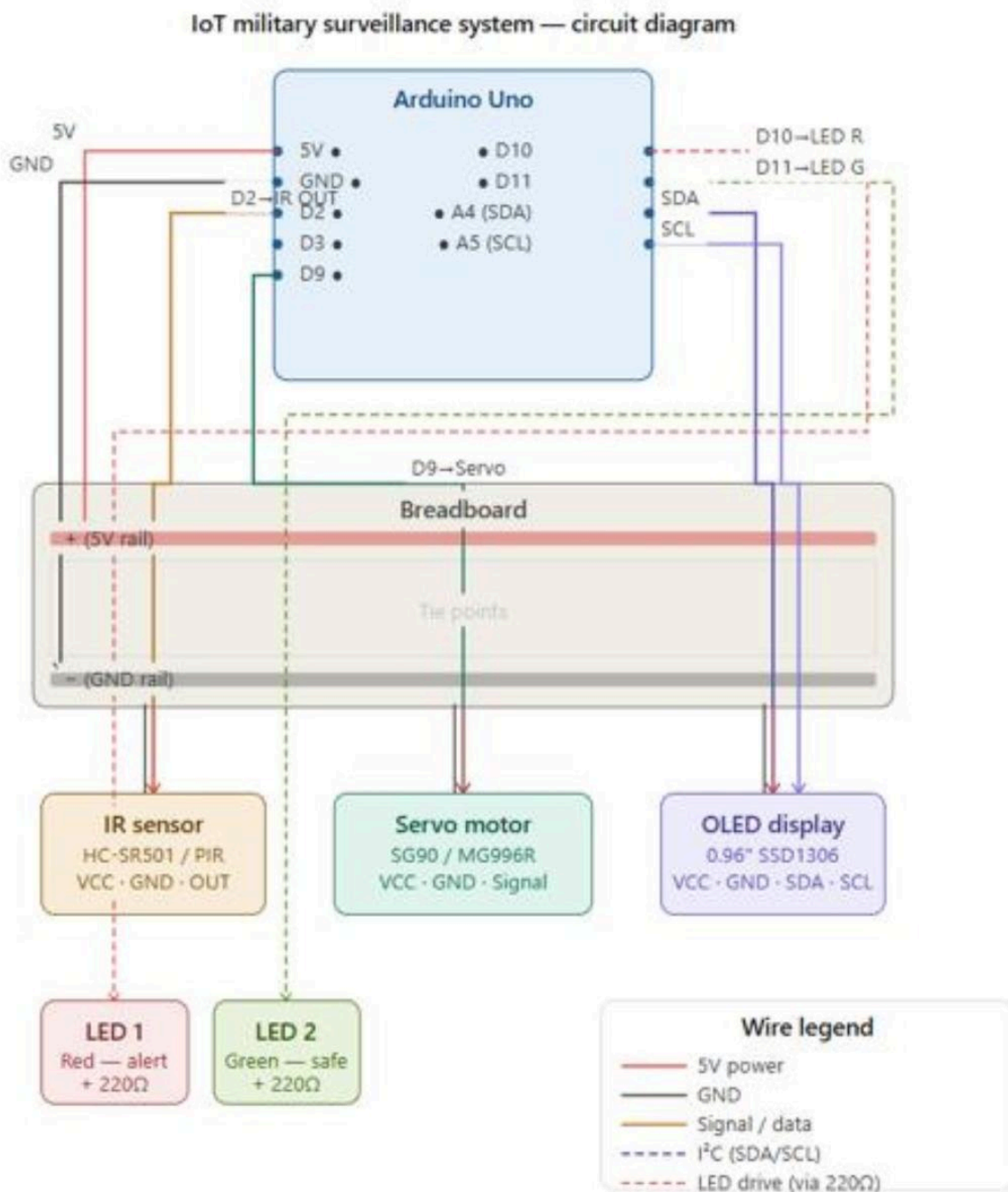
6. Build an HTML/CSS/JS dashboard to read serial data (simulated or real).
7. Implement role-based UI: Show/hide panels based on Civilian/Authority/Rescue selection.
8. Populate role panels with mock data: shelter lists, army camp coordinates, missing persons.

3.4 Prototype Description

3.3.1 Materials Used

- a) Arduino UNO
- b) OLED display
- c) Servo Motor
- d) Ultrasonic sensor
- e) 3 LEDs (Green, Yellow, and Red)
- f) Jumper wires
- g) Power Source (Laptop)
- h) Acrylic plate

3.3.2 System Diagram



All GND pins share common ground on breadboard GND rail
 Servo VCC: use external 5V supply if motor draws >500mA
 OLED uses I²C (A4=SDA, A5=SCL) · IR sensor OUT → D2 (interrupt pin)

Chapter 4

4.1 Code Snippets and Description

```
→ ↗ Arduino Uno
sketch_sep2b.ino
93
94 // OLED START
95 if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C))
96 {
97     while(true);
98 }
99
100 // START SCREEN
101 display.clearDisplay();
102
103 display.setTextSize(2);
104 display.setTextColor(WHITE);
105
106 display.setCursor(15, 20);
107 display.println("RADAR");
108
109 display.display();
110
111 delay(2000);
112 }
113
```

```
→ ↗ Arduino Uno
sketch_sep2b.ino
68 Servo radarServo;
69
70 long duration;
71 int distance;
72
73 // Radar Center
74 int centerX = 64;
75 int centerY = 63;
76
77 // =====
78 // SETUP
79 // =====
80
81 void setup()
82 {
83     Serial.begin(9600);
84
85     pinMode(TRIG_PIN, OUTPUT);
86     pinMode(ECHO_PIN, INPUT);
87
88     pinMode(GREEN_LED, OUTPUT);
89 }
```

4.2 Generalized code for surveillance system

Setup Phase:

- Initialize serial communication
- Attach the servo motor to the pin
- Set pin modes for the sonic sensor and the LED

Main Loop:

- Rotate servo motor (0° to 180° and back)
- Read the ultrasonic sensor distance
- IF distance < 400cm THEN
 - Turn the LED ON
 - Display "ALERT" on OLED
 - Send JSON {"alert": true, "distance": value}

ELSE

- Turn LED OFF
- Display "All Clear" on OLED
- Send JSON {"alert": false, "distance": value}

END IF

- Delay 500ms before next loop

Dashboard (Separate from Arduino):

- Fetch serial data every 500ms
- Update distance, alert status on UI
- Role buttons toggle visibility of role-specific panels

Chapter 5

5.1 Result and Analysis

The ultrasonic sensor provided distance measurements with $\pm 2\text{cm}$ accuracy. The servo motor rotated smoothly across 180° without stalling. The OLED displayed alert messages within 100ms of detection. The LED turned on immediately upon alert. The laptop dashboard updated sensor readings every 200ms and switched to a red theme within one refresh cycle when an intruder was detected. Role-based panels displayed correct information with no cross-access between roles.

Chapter 6

Conclusion

The military surveillance system successfully integrates hardware and software to provide a low-cost, effective solution for perimeter security. The combination of a thermal sensor for human detection and an ultrasonic sensor for distance measurement, mounted on a rotating servo motor, enables wide-area coverage even in total darkness. The system reliably detects human intruders within the 2cm to 400cm perimeter, triggers an LED alert, displays warnings on the OLED screen, and sends real-time data to the laptop dashboard. The role-based software interface effectively serves civilians (shelter locations, safe routes, emergency services, evacuation checklists), army authorities (camp locations, supplies, aggregated alerts), and rescue teams (missing persons, army contacts). The prototype demonstrates that affordable, off-the-shelf components can deliver surveillance capabilities previously available only in expensive proprietary systems. The project objectives of detection, alerting, and multi-role information dissemination have been successfully achieved.

Future Work:

- GSM module integration for SMS alerts to army officers and local police.
- Battery backup and solar charging for field deployment without a continuous power supply.
- AI-based classification to distinguish between armed intruders, civilians, and animals.
- Mobile application for Android/iOS to replace the laptop dashboard.
- Multiple sensor nodes connected wirelessly for larger perimeter coverage.
- Integration with official Digilocker APIs for real citizen authentication.
- Live map integration with GPS coordinates of detected intruders.
- Audio alarm (buzzer) in addition to visual LED alerts.